

Tutorial 10

Linear transformations

1. Which of the following functions are linear? Give reasons for your answers.

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (x + y, 2x - y - 3).$$

(b) $T : M_{3 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \text{trace}(A)$.

(If you have forgotten what the trace of a square matrix is, look at Question 3 in Tutorial 9.)

(c) $T : M_{3 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \det(A)$.

(d) $R : C[0, 1] \rightarrow \mathbb{R}^2$ defined by

$$R(f) = (f(0), \int_0^1 f(x) dx).$$

(Recall that $C[0, 1]$ is the vector space of all real-valued continuous functions defined on the interval $[0, 1]$.)

2. (a) Show that

$$B = \{1 + x, 1 + x^2, x + x^2\}$$

is a basis for $P_2(x)$.

(b) It is given that $T : P_2(x) \rightarrow P_1(x)$ is a linear transformation and that

$$T(1 + x) = 2 - x, \quad T(1 + x^2) = 3x, \quad T(x + x^2) = 4 + x.$$

Find $T(2 - x^2)$.

(c) For $a, b, c \in \mathbb{R}$, find $T(a + bx + cx^2)$.

(d) Find the range of T .

(e) Is $\{T(1 + x), T(1 + x^2), T(x + x^2)\}$ a basis for the range of T ? Give a reason for your answer.

3. Let the function $T : M_{2 \times 2}(\mathbb{R}) \rightarrow P_2$ be defined by

$$T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a+b)x^2 + (b+c-d)x + (c-d).$$

- (a) A student says: “It is easy to see that T is a linear transformation; it follows from the fact that $T(\mathbf{0}) = \mathbf{0}$.” What is wrong with this argument?
 - (b) Show that T is linear.
 - (c) Recall that the kernel of T is given by $\ker(T) = \{A \in M_{2 \times 2}(\mathbb{R}) : T(A) = \mathbf{0}\}$. Find $\ker(T)$.
 - (d) Find a basis for the kernel of T . What is $\dim(\ker(T))$?
 - (e) Is T one-to-one? Give a reason for your answer.
 - (f) Recall that the range of T is given by $\text{ran}(T) = \{T(A) : A \in M_{2 \times 2}(\mathbb{R})\}$. Find $\text{ran}(T)$.
 - (g) Find a basis for the range of T . What is $\dim(\text{ran}(T))$.
 - (h) Is T an onto function (surjective)? Give a reason for your answer.
4. Let V be the vector space of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$. The function $T : V \rightarrow \mathbb{R}^2$ is defined by:

$$T(f) = (f(0), f'(0)), \quad \text{where } f \in V.$$

- (a) What is $T(f)$ when $f(x) = cx + d$?
- (b) Is T a linear map? Give reasons for your answer.
- (c) It is possible to describe the functions in $\ker(T)$ in terms of their graphs. Complete the following description:
“A function is in $\ker(T)$ if and only if its graph ...”
- (d) Is T one-to-one (injective)? Give reasons for your answer.
- (e) Find the range of T . Is T onto (surjective)?

5. Let V and W be vector spaces, $T : V \rightarrow W$ a linear map and $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for V . Put $C = \{T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)\}$.
- Prove that C spans the range of T .
[Hint: If $\mathbf{v} \in \text{ran}(T)$, then there is a $\mathbf{u} \in V$ such that $\mathbf{v} = T(\mathbf{u})$.]
 - Show that C is not necessarily linearly independent.
[Hint: Give an example of a T and a B for which C is not linearly independent. Question 2 may help you here.]
 - What property must T have to ensure that C will be linearly independent? Give a proof for your answer.
[Hint: Start with $\lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_n T(\mathbf{b}_n) = \mathbf{0}$ and then use the fact that T is linear to write the left hand side in a different way.]
 - What property must T have to ensure that C will be a basis for $\text{ran}(T)$?
6. In this question you have to use your knowledge of bases and coordinate vectors to prove something about a function between two vector spaces.
- Let V be an n -dimensional real vector space and $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ an ordered basis for V . If $\mathbf{v} \in V$, we write $[\mathbf{v}]_B$ for the coordinate vector of $\mathbf{v}$ with respect to the basis B .
- Prove that if $\mathbf{u}, \mathbf{v} \in V, \lambda \in \mathbb{R}$, then $[\lambda \mathbf{u}]_B = \lambda [\mathbf{u}]_B$ and $[\mathbf{u} + \mathbf{v}]_B = [\mathbf{u}]_B + [\mathbf{v}]_B$.
[Hint: There is a unique way of writing $\mathbf{u}$ as a linear combination of elements of B ; the scalars used give you $[\mathbf{u}]_B$. To find $[\lambda \mathbf{u}]_B$, you need to express $\lambda \mathbf{u}$ as a linear combination of elements of B ; there is only one way of doing it. Find it! The same procedure works for $[\mathbf{u} + \mathbf{v}]_B$.]
 - Let the function $F : V \rightarrow \mathbb{R}^n$ be defined by $F(\mathbf{v}) = [\mathbf{v}]_B$. What does (a) tell you about the function F ?
 - Is the function F defined in (b) one-to-one? Give a reason for your answer.
 - Is $\{[\mathbf{b}_1]_B, [\mathbf{b}_2]_B, \dots, [\mathbf{b}_n]_B\}$ a basis for $\mathbb{R}^n$? Give a reason for your answer.